Cloud Computing Mini Project Reflection

Roll number: Not provided

This project showed that an end-to-end machine-learning service depends on more than model training. The teacher worker needed bounded prompts, timeouts, and reliable JSON cleaning before it could produce usable data. The 50-record subset exposed how sensitive a 70M parameter student is to data quality and scale: teacher outputs were usable, but the student produced invalid frames and would not meet a production quality gate.

The Kubernetes phase also demonstrated the importance of operational diagnostics. Node disk pressure, local image imports, model startup, DNS, SeaweedFS routing, and pod scheduling each created distinct failures. Pinning the image to the node containing it and replacing the external SeaweedFS DNS dependency with a stable cluster service and endpoint made the deployment reproducible.

The load and recovery test proved that rolling updates and pod replacement can be verified with headers, readiness, and traffic. The v2 rollout was observed with X-Model-Version=v2, then rolled back to v1. I would improve the next iteration by generating more teacher records, retraining on a larger balanced subset, adding automated model readiness timeouts, and keeping quality gates mandatory for release promotion.